

Accuracy of the Scaled Flux Method for Spatially Resolved Nuclide Transmutation in Flowing-Fuel Molten Salt Reactors

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Abstract

Depletion simulations of flowing fuel molten salt reactors do not typically distinguish between in-core and ex-core regions. Instead, they either irradiate the entire primary loop while scaling the flux, or irradiate the in-core volume alone, which leaves some amount of fuel salt out of the simulation. Either approximation results in some level of inaccuracy in nuclide concentrations. The method of scaling the flux is more commonly used, but this approach assumes perfect mixing of the fuel salt in the primary loop. This work investigates a spatially resolved method to account for the impact flow and chemical removal has on concentrations of nuclides rapidly and accurately. Specifically, the spatially resolved model uses one-dimensional advection, which assumes no mixing in the primary loop. This model is used to provide the most conservative possible comparison, as any amount of mixing would only move the result closer to the scaled flux method. Comparisons are made in this work between this spatially resolved model, scaled flux model, other work from the literature, and experimental data. These comparisons show that the scaled flux method is sufficient for modeling most nuclides unless there is a large spatially dependent loss term, near-stationary flow rate, or artificial power oscillations with almost no mixing. In such cases, a spatially resolved method is necessary.

Keywords — Molten Salt Reactor, Molten Salt Reactor Experiment, Depletion, Chemical Removal, Flow Rate, Power Oscillations

I. INTRODUCTION

Liquid-fueled molten salt reactor (MSR) development started in the 1950s with the Aircraft Reactor Experiment [1]. Progress continued in the Molten Salt Reactor Experiment (MSRE) in the 1960s [2]. After that, solid-fueled light water reactors became the focus, which have since become the most common nuclear reactor used in the industry. However, the Generation IV International Forum in the early 2000s designated the MSR as one of the six advanced reactors of interest moving forward [3]. Since then, MSRs have garnered further interest, such as the 2020 Advanced Reactor Demonstration Program award for \$113 million over seven years to the Molten Chloride Reactor Experiment [4].

Interest has shifted to MSRs, and many existing simulation codes and methods must adjust to account for the moving fuel. In particular, depletion simulations are of interest. In solid-fueled reactors, the fuel does not experience significant advection or diffusion of nuclides. However, MSRs have the fuel circulate through the in-core and ex-core regions. The depletion simulations must account for these regions with different power densities and the effects of circulation. Modeling this behavior accurately is vital, as the fuel composition affects reactor dynamics, reactor safety, and safeguards.

One approach is computationally inexpensive and is called the scaled flux method [5]. The scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. This approach does not model the in-core and ex-core regions explicitly, and instead assumes the salt is perfectly mixed in the primary loop.

Another approach has shown how spatial resolution can be investigated using a system of ordinary differential equations to solve the Bateman equation. This approach can estimate the steady-state concentration of certain nuclides across a flow loop, but it does not account for temporal resolution [6].

The toolkit MCNP/ADDER [7] has also been used to account for flow and chemical removal between neutron transport-coupled fuel depletion steps over the MSRE power history [8]. But within ADDER, ex-core loss terms (e.g., decay and chemical removal) are only treated after the calculation of the coarse depletion steps (e.g., months), as opposed to simultaneously on the timescale of the flow loop residence time (e.g., seconds). This results in an accurate removal of elements except for those nuclides that have both an ex-core loss term and a non-negligible neutron

absorption cross section.

Alternatively, a more accurate treatment was recently demonstrated through MOOSE-based coupling of the deterministic neutron transport code Griffin with the thermal fluids code Pronghorn, allowing for spatial and temporal resolution of certain nuclides [9]. Tano et al. present an approach where only insoluble, medium-lived, and high concentration nuclides are tracked with pre-computed spatial distributions, which only need to be updated if the medium-lived isotopes are not in equilibrium after the thermal-hydraulics perturbation.

Spatially and temporally resolved approaches, such as the one used by Tano et al., are more computationally expensive than those that only require solving a system of ordinary differential equations. However, Tano et al. has shown that including circulation of the fuel salt in depletion directly affects the equilibrium concentration of ^{135}Xe by approximately 45% compared to a spatially unresolved approach without flux scaling [9]. This indicates there may be a need for spatially resolved depletion simulations to get accurate fuel salt compositions.

Tano et al. and Betzler have both described fully modeling advection in depletion as computationally intractable [9, 10]. This is because tracking all nuclides over space and time requires far more computational work than handling a traditional time dependent system of ordinary differential equations (ODEs). This spatially resolved system of partial differential equations (PDEs) requires accuracy in the short time scale flow effects. These small time steps cause multi-year depletion simulations to be exceedingly expensive.

However, Tano et al. showed that not all nuclides must be spatially resolved to obtain accurate results [9]. The present work utilizes this idea from Tano et al. and extends a previous steady-state approach [6]. This work isolates isobars of interest and adjusts the steady-state approach by accounting for time resolution in the system of differential equations. High-fidelity thermal hydraulic simulations are computationally expensive, so instead, a spatially resolved model with 1D advection is used to capture the in-core and ex-core effects. With the simplifications implemented, this method can run on ~~an~~ a laptop in minutes, rather than requiring a supercomputer allocation.

Because this model does not account for any mixing, it will provide the largest difference from the scaled flux method, which uses perfect mixing. If the scaled flux model and the spatially resolved model with 1D advection give the same results, then that means that both no mixing and perfect mixing will give the same results for that problem. This means that, for that category of

problem, the scaled flux method can be used over a spatially resolved method with low cost and high accuracy.

II. METHODS

The ODE form of the Bateman equation written in simplified terms for nuclide i , which only contains time dependence, is given as

$$\frac{dN_i(t)}{dt} = S_i(t) - \mu_i(t)N_i(t), \quad (1)$$

$$S_i(t) = \phi(t)FY_i\Sigma_f f_{in} + \sum_{j \neq i} \lambda_j N_j(t) \gamma_{j \rightarrow i}, \quad (2)$$

$$\mu_i(t) = \lambda_i + r_i(t)f_{ex} + \phi(t)\sigma_{a,i}f_{in}, \quad (3)$$

where

$$\begin{aligned} N_i &= \text{Concentration of nuclide } i \text{ [atoms} \cdot \text{cm}^{-3}] \\ S_i &= \text{Source rate [atoms} \cdot \text{cm}^{-3} \cdot \text{s}^{-1}] \\ \mu_i &= \text{Loss rate [s}^{-1}] \\ \phi(t) &= \text{Time dependent neutron flux [n} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}] \\ FY_i &= \text{Fission yield of nuclide } i \text{ [-]} \\ \Sigma_f &= \text{Macroscopic fission cross section of the fuel [cm}^{-1}] \\ \gamma_{j \rightarrow i} &= \text{Branching ratio of } j \text{ to } i \text{ [-]} \\ \lambda_j &= \text{Decay constant of nuclide } j \text{ [s}^{-1}] \\ r_i &= \text{Chemical removal rate [s}^{-1}] \\ \sigma_{a,i} &= \text{Microscopic absorption cross section of nuclide } i \text{ [cm}^2] \\ f_{in} &= \text{Fraction of fuel salt in-core [-]} \\ f_{ex} &= \text{Fraction of fuel salt ex-core [-].} \end{aligned} \quad (4)$$

The unscaled flux method uses Equations (1), (2), and (3) where the in-core and ex-core fractions of fuel salt, f_{in} and f_{ex} , are equal to one. This is because chemical removal is applied

to the entire salt volume unscaled, as is the neutron flux. The scaled flux method uses the same equation, but it applies the in-core and ex-core fractions appropriately. When one-dimensional spatial dependence is added to Equation (1) with only advection considered, a PDE is created of the form

$$\frac{\partial N_i(z, t)}{\partial t} + \nu(z, t) \frac{\partial N_i(z, t)}{\partial z} = S_i(z, t) - \mu_i(z, t) N_i(z, t), \quad (5)$$

where

$$\begin{aligned} \nu(z, t) &= \text{linear flow rate } [cm \cdot s^{-1}] \\ z &= \text{linear distance through primary loop } [cm]. \end{aligned} \quad (6)$$

Equation (5) is used in this work with a periodic boundary condition to provide the spatially resolved concentrations for target nuclides. The source term is comprised of decay sources and fission yields in this work. Sources of nuclide i from parasitic absorption are currently not considered, as the impact is assumed to be sufficiently smaller than the fission and decay parent sources. The loss term includes losses from both decay and the ~~neutron total~~ total neutron capture cross section in the core region. The loss term can also include a chemical removal term resolved in both space and time for nuclide i .

To solve the coupled system of equations given in Equation (5) in a computationally efficient manner, several simplifications are made. The results do not account for changes in the fuel or evolution of the neutron energy spectra [11]. Instead, these terms are assumed to be constant, which can be a reasonable approximation if fresh fuel is added and fission product poisons are removed during online reprocessing. Another simplification in this work is the use of only one spatial dimension and uniform flow. This includes simplifications to the flow of the fuel salt, which can only change in time and the single spatial dimension.

II.A. Data

Custom scripts combined with OpenMC use the linear flow rate, average neutron energy, average neutron flux, in-core fraction of fuel salt, ex-core fraction of fuel salt, in-core fuel salt volume, ex-core fuel salt volume, and average temperature to generate the appropriate cross sections and

terms in Equation (5). The solver uses the average temperature and energy to generate one-group cross sections from OpenMC.

The scripts use OpenMC to access general nuclear data in ENDF-B/VII.1 [11, 12]. These data include fission yield fractions, decay chains, branching ratios, half-lives, and cross sections. These data are important for properly constructing the source and loss terms for each nuclide, while the decay chains are useful for constructing the relationships between them.

The results in this work use the data shown in Table I. The fissile concentration, neutron flux, nominal power, power history, and in-core fraction come from the MCNP/ADDER simulation of the MSRE [7, 8]. The other terms come from the data from the OpenMSR project [13].

TABLE I
Simulation parameters used in this work.

Parameter	Value	Units
Fissile Nuclide	^{235}U	-
Fissile Concentration	8.41×10^{19}	$\frac{\text{atoms}}{\text{cm}^3}$
Neutron Flux	8.4×10^{12}	$\frac{n}{\text{cm}^2\text{s}}$
Neutron Energy	0.02533	eV
Salt Density	2.32556	$\frac{g}{\text{cm}^3}$
Temperature	294 900	K
Nominal Power	7.34	MW
Primary Loop Flow Length	608.06	cm
In-core Salt Fraction	27.2	%
Ex-core Salt Fraction	72.8	%
Volume	2.116111	m^3
Linear Flow Rate	21.75	$\frac{\text{cm}}{\text{s}}$

Figure 1 shows the flow path along which the spatial nodes are distributed. Most of the path is in the ex-core region, but 27.2% is in the in-core region, which takes up 165.39 centimeters of the 608.06 centimeter flow path length.

II.B. Solver

This work uses finite differencing upwind, or backwards, in space and explicit in time to generate a coupled system of PDEs. Equation (7) is used to solve for the concentration at each subsequent time step as:

$$N_k^{(l+1)} = N_k^{(l)} + \Delta t \left(S_k^{(l)} - \mu_k^{(l)} N_k^{(l)} - \nu_k^{(l)} \frac{N_k^{(l)} - N_{k-1}^{(l)}}{\Delta z} \right), \quad (7)$$

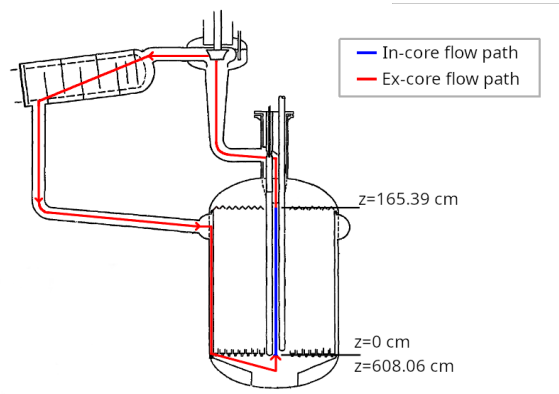


Fig. 1. [The flow path and bounds for the in-core region of the MSRE, overlayed on the MSRE loop diagram from \[14\].](#)

where

$$\begin{aligned}
 N_k^{(l)} &= \text{Concentration at node } k \text{ and time } l \text{ [atoms} \cdot \text{cm}^{-3}] \\
 \mu_k^{(l)} &= \text{Loss term at node } k \text{ and time } l \text{ [s}^{-1}] \\
 S_k^{(l)} &= \text{Source term at node } k \text{ and time } l \text{ [atoms} \cdot \text{cm}^{-3} \cdot \text{s}^{-1}] \\
 \nu_k^{(l)} &= \text{Flow rate at node } k \text{ and time } l \text{ [cm} \cdot \text{s}^{-1}].
 \end{aligned} \tag{8}$$

The upwind in space approach is used to maintain numerical stability [15]. This is investigated more thoroughly in Section III.C, which investigates the sensitivity of the solution to the Courant number, given in Equation (9) as:

$$C = \frac{\Delta t \nu}{\Delta z}, \tag{9}$$

where

$$\begin{aligned}
 \Delta t &= \text{time step [s]} \\
 \Delta z &= \text{spatial grid size [cm]}.
 \end{aligned} \tag{10}$$

This is important as it is used for the Courant–Friedrichs–Lewy condition, where the Courant number, C , must be less than or equal to 1 for the explicit finite differencing scheme presented in this work. Equation (9) shows that, as the spatial mesh, Δz , becomes finer or the flow rate, ν ,

becomes larger, the time step, Δt , must become smaller, which means higher computational cost. This is because a smaller time step becomes necessary to capture the effects spatially when the fluid moves faster through a spatial region Δz .

II.C. Evaluation Criteria

The spatially resolved model provides a concentration profile at each point in time, while the scaled flux model provides a single concentration for each nuclide at each point in time. By comparing Equation (1) and (5), it becomes noticeable that, aside from the spatial dependence, the main difference is the existence of the advection term. As the advection term goes to zero, which could happen by the linear flow rate or the concentration gradient going to zero, then the equations closely resemble each other. In such a case, when Equation (5) is integrated over space, then the equations are equal.

Based on this observation, the metric selected to observe is the difference of each nuclide, given in Equation (11) as:

$$d_i(t) = \left| \frac{\bar{N}_{1D,i}(t) - N_{0D,i}(t)}{\bar{N}_{1D,i}(t)} \right|, \quad (11)$$

where

$$\begin{aligned} d_i(t) &= \text{Difference metric [-]} \\ \bar{N}_{1D,i}(t) &= \text{Spatially averaged spatially resolved model concentration of nuclide } i \text{ [atoms} \cdot \text{cm}^{-3}] \\ N_{0D,i}(t) &= \text{Scaled or unscaled flux model concentration of nuclide } i \text{ [atoms} \cdot \text{cm}^{-3}]. \end{aligned} \quad (12)$$

When the difference metric approaches zero, that means that the perfect mixing model and the no mixing model give the same result. If the difference metric is non-zero, that means that the spatially resolved model has a non-zero result for the advection term. For any intermediate scenario, such as inclusion of diffusion or turbulent mixing, the result can be expected to lie somewhere between the perfect mixing and no mixing results. Thus, the larger the difference metric, the greater the thermal-hydraulic modeling impact will be for that specific problem.

III. RESULTS

The results for this work include various sensitivity studies, a comparison between spatially resolved and spatially unresolved approaches, validation, and code-to-code benchmarking. Most of the results focus on ^{135}Xe , though the validation uses ^{95}Nb and ^{95}Zr . ^{135}Xe is selected for the sensitivity studies for several reasons: it has a large parasitic absorption cross section, which makes it likely to have a larger spatial dependence than other nuclides; it is important for modeling reactor behavior due to its high absorption cross section, which means evaluating it is useful; and it has a non-zero chemical removal rate, which means the chemical removal rate sensitivity can be compared with removal rates from the literature.

III.A. Spatial Sensitivity

The spatial sensitivity study informs the other analyses what level of spatial resolution is required to generate sufficiently accurate results. The time stepping sensitivity is built-in with this study, as a finer spatial mesh also requires a finer time stepping scheme due to preserving a constant Courant number of ~~0.5~~1.0, as shown in Equation (9). The choice of Courant number is discussed further in Section III.C. This analysis compares the concentration of ^{135}Xe after one hour in the MSRE using the spatially resolved PDE method for various numbers of spatial nodes. Table II provides details on the cost and accuracy of each simulation.

TABLE II

Spatial sensitivity detailing how cost and accuracy vary based on the number of spatial nodes for ^{135}Xe with a single tracked nuclide and a Courant number of ~~0.5~~1.0 over one hour in the MSRE. The number of nodes selected for generating results is shown in bold.

Nodes	Cost [s]	Error [%]
50	0.28 <u>0.14</u>	1.81 <u>2.59</u>
100	0.84 <u>0.42</u>	1.26 <u>1.25</u>
200	2.76 <u>1.29</u>	0.82
500 500	16.03 6.93	0.11 0.11
1000	60.71 <u>26.63</u>	0.06
2000	224.43 <u>103.12</u>	0.01
4000	884.60 <u>413.30</u>	-

The computational cost scales approximately as $O(n^2)$ with the number of spatial nodes, n . Based on these results, to ensure that any small difference metric values are detectable, 500 nodes are used. This ~~neecessitates an increased computational cost, but~~ number of nodes allows for

analysis of differences with ~~uncertainty~~ uncertainties of approximately 0.1%.

III.B. Chain Sensitivity

The chain sensitivity study informs the other analyses what number of nuclides to include in the decay chain. This study compares the calculated ^{135}Xe concentration after one hour for various numbers of decay parents for ^{135}Xe included in the decay chain. Table III shows how varying the number of decay parents for ^{135}Xe in the isobar affects the results and simulation time.

TABLE III

Chain sensitivity detailing how cost and accuracy vary based on the number of nuclides for ^{135}Xe with 500 spatial nodes and a Courant number of ~~0.5~~ 1.0 over one hour in the MSRE. The nuclides selected for generating results are shown in bold.

Nuclides	Cost [s]	Error [%]
Xe	14.17 <u>7.40</u>	84.07 <u>84.05</u>
Xe, ^mXe	28.85 <u>15.01</u>	58.39 <u>58.38</u>
Xe, ^mXe , I	42.56 <u>21.27</u>	31.75 <u>31.73</u>
Xe, ^mXe , I, Te	58.49 <u>28.10</u>	1.08
Xe, ^mXe, I, Te, Sb Xe, ^mXe, I, Te, Sb	74.03 <u>35.30</u>	0.00 <u>0.00</u>
Xe, ^mXe , I, Te, Sb, Sn	83.63 <u>42.14</u>	-

The cost scales as $O(n)$ for n nuclides. Five and six nuclides have the same error because ^{135}Sn does not decay into ^{135}Sb in the OpenMC chain file used, so not including ^{135}Sn yields the same result. The expected contribution of ^{135}Sn is small regardless, so the error would also be small if it were included. Using five nuclides in the 135 isobar leading to ^{135}Xe yields accurate results, so this number of nuclides is used for the other ^{135}Xe analyses. For other analyses not using ^{135}Xe , since the nuclide cost scaling is only linear, six nuclides are used for ^{95}Zr and seven for ^{95}Nb .

III.C. Courant Number Sensitivity

This sensitivity study investigates the time step size represented by the Courant ~~, or CFL,~~ number, used in the Courant–Friedrichs–Lewy convergence condition. A constant flow rate is used, as well as a constant number of nodes as described in Section III.A. There are two different studies used to evaluate varying the value of the Courant number. The first is a one minute MSRE simulation with 500 spatial nodes and 5 nuclides leading to ^{135}Xe . The results of this study are shown in Table IV. The results in this table show that the cost scales approximately as $O(n)$, where

n is the number of time steps. These results also show that, at one minute, there is no monotonic correlation between the Courant number and the percent difference from the concentration of ^{135}Xe with a Courant number of 0.1. Beyond a Courant number of 1.0, the solution becomes non-physical.

TABLE IV

Courant number sensitivity study on the concentration of ^{135}Xe with five nuclides in the MSRE based on the time step size used with 500 spatial nodes over the course of one minute. The Courant number selected for generating results is shown in bold.

Courant Number	Time Steps	Cost [s]	Error-Difference [%]
1.0 1.0	2994 1073	1.59 0.06	0.44 7.01
0.9	3327 1192	1.77 0.71	0.48 8.03
0.8	3743 1341	1.98 0.74	0.31 2.99
0.7	4278 1532	2.27 0.84	0.38 1.98
0.6	4990 1788	2.65 0.98	0.17 5.03
0.5	5988 2146	3.19 1.22	0.28 2.00
0.4	7484 2682	4.22 1.46	0.17 3.02
0.3	9980 3576	5.65 1.97	0.07 2.02
0.2	14970 5365	8.02 2.92	0.03 1.01
0.1	29940 10730	16.03 5.90	-

The next study to investigate is a longer, one hour MSRE simulation with the other parameters unchanged from the previous study. The results of this study are shown in Table V. ~~The Courant number changes between these studies because the time step is calculated by dividing the increased total time by the unchanged number of time steps~~ Once again, these results show that, at one hour, there is no monotonic correlation between the Courant number and the percent difference from the concentration of ^{135}Xe with a Courant number of 0.1.

~~From these studies~~ At both one minute and one hour, the error seems to somewhat follow the number of time steps and depends less on the Courant number, though the trend is not monotonic. There also appears to be an inverse correlation of the ~~net simulation time~~ simulated time in the MSRE with the error for a given number of time steps, as the error for a one hour simulation was lower in comparison to all the one minute simulations with the same Courant number.

These trends are understood more clearly when investigating the effect of varying the Courant number on the simulation. The non-monotonic variation in error with varying Courant number arises from the competing influences of numerical diffusion and phase accuracy introduced by changes in the time step. The phase of the low Courant number result is more accurate while the diffusion of the high Courant number result is more accurate. However, the phase of the low

TABLE V

Courant number sensitivity study on the concentration of ^{135}Xe with five nuclides in the MSRE based on the time step size used with 500 spatial nodes over the course of one hour. The Courant number selected for generating results is shown in bold.

Courant Number	Time Steps	Cost [s]	Error Difference [%]
1.0 1.0	179640 64384	95.26 35.26	0.0084 0.21
0.9	199600 71538	107.23 40.51	0.0075 0.08
0.8	224550 80481	119.41 44.24	0.0066 0.16
0.7	256629 91978	136.23 52.29	0.0112 0.13
0.6	299400 107308	160.56 58.86	0.0047 0.00
0.5	359280 128769	196.97 70.45	0.0037 0.08
0.4	449100 160962	254.26 89.20	0.0028 0.05
0.3	598800 214616	332.39 119.76	0.0019 0.00
0.2	898200 321924	477.75 179.72	0.0009 0.00
0.1	1796400 643848	956.84 349.55	-

Courant number can be fixed by adjusting the advection rate, which is as demonstrated in Figure 2.

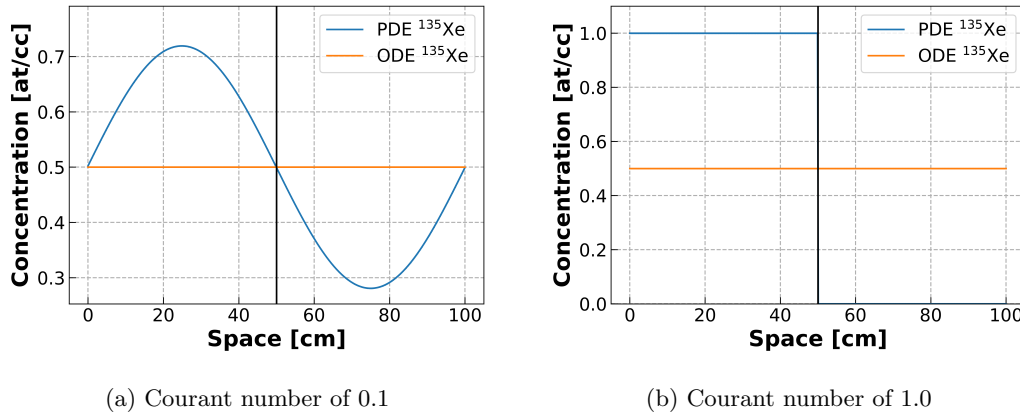


Fig. 2. Concentration for a simple test case in a 100 *cm* long channel with a flow rate of 10 *cm/s* after 5 minutes with no source or loss terms and an initial concentration in the first 50% of the loop.

~~Based on this analysis~~In Figure 2(a), the Courant number ~~should be selected based on the~~ desired behavior, particularly since the effect on the percent difference is less than half a percent ~~for all studied cases~~of 0.1 has an accurate phase, but is highly diffusive. Because the ~~evaluation~~ criteria depends on the spatial gradient, preserving the sharpness of the gradient is deemed to be the most important. ~~This means a purpose of only using advection is to capture the difference between models with no mixing and perfect mixing, this model is not useful due to mixing by numerical diffusion.~~ Figure 2(b), the Courant number of ~~one is optimal for the analyses studied.~~

~~Additionally~~ 1.0 has a slightly inaccurate phase, but does not have numerical diffusion. However, the phase shift can be ~~corrected in a straightforward manner by adjusting the advection rate~~ accounted for by scaling the salt advection speed by a constant, which removes this drawback, as shown in Figure 2(b). For this work, the advection speed was found to be most accurate when multiplied by a factor of 1.002 for each case studied. ~~This means a Courant number of one is optimal for the analyses studied.~~

III.D. Validation and Code-to-Code Benchmarking

This work uses the experimentally measured MSRE ladle salt samples over time during uranium-235 operation for validation [16]. The validation uses concentrations for ^{95}Zr and ^{95}Nb during operation in the MSRE from January 24th, 1966 to November 7th, 1967. ^{95}Zr is a salt-seeking nuclide without an appreciable noble gas precursor, which means it can be modeled without having to account for chemical removal. The ^{95}Nb acts like a noble metal in the fuel salt, so it has some chemical removal rate due to plate-out in the heat exchanger. However, fully understanding the behavior of ^{95}Nb from the existing data is challenging [17]. For this benchmark, no chemical removal is included, though there are rates that exist in the literature for modeling removal from the fuel salt to the cover gas in the pump bowl [18].

Figure 3 shows the concentrations of ^{95}Zr and ^{95}Nb over time. In the figure, comparisons are made between the experimental results of the MSRE [16], analysis using MCNP/ADDER [8], and the current work using the spatially resolved and scaled flux models. There appears to be reasonable agreement between the current work and the results from MCNP/ADDER. This agreement shows that the six to seven nuclides are sufficient to get a reasonable concentration for a specific target nuclide compared to a full depletion matrix. ~~There is also agreement between the spatially resolved and scaled flux concentrations. Both models are overlapping on these plots because the simulation of the MSRE during this time frame does not have any properties which induce error in the scaled flux method. This is discussed in more detail in Section III.E.~~

However, the MSRE experimental data does not agree at each point, particularly for ^{95}Nb . The ^{95}Nb concentrations from the MSRE measurements are smaller than the model predicts, which means that a non-zero chemical removal rate would likely fit these data better. This lack of agreement ~~for ^{95}Zr and ^{95}Nb~~ can be explained by uncertainty in the measurements, chemical

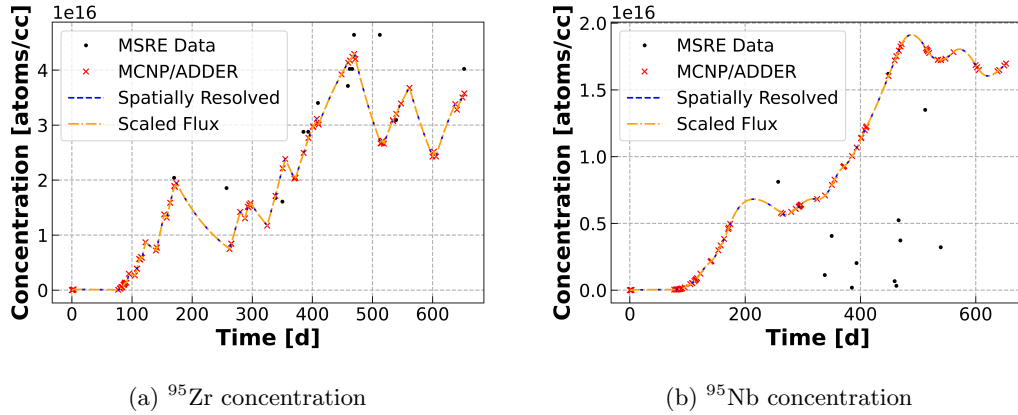


Fig. 3. Concentrations over time in the Molten Salt Reactor Experiment with no chemical removal.

effects which aren't modeled in MCNP/ADDER or this work, and other ~~minor~~-contributing factors these models do not include. ~~Particularly, the~~

~~The MSRE experimental data for ^{95}Nb concentrations from the MSRE measurements are smaller than the model predicts, which means that a non-zero chemical removal rate would likely fit these data better~~ ^{95}Zr agrees reasonably well with the models presented in this work, as the points generally fall on or near the modeled concentration. These models are not expected to perfectly capture the physics of the MSRE, but rather the general trend of the evolution of the nuclide concentrations. Thus, the general agreement with ^{95}Zr demonstrates the models are successfully capturing this general trend.

Comparison between ^{135}Xe chain concentrations are made with a steady-state radiochemical transport model for additional code-to-code benchmarking [6]. ~~Two variations are compared, one with nine regions and one with two regions~~ The steady-state model uses in-house Python scripts and the same starting equation shown in Equation (5). The MSRE is discretized into either nine or two regions, with flow paths between them to model the salt flow path. The scripts sets the derivative of the concentration with respect to time equal to zero, turning the coupled system of PDEs into a system of ODEs, with one ODE per region.

The code-to-code benchmarking compares the nine region and two region steady-state models, shown in Table VI. The input data for the steady-state radiochemical model was slightly modified to match Table I. However, one difference that is likely to lead to discrepancies is the treatment of ^{135m}Xe . In this work, the source of meta-stable xenon includes an independent fission yield and

from decay of iodine. In the steady-state radiochemical transport model, a cumulative fission yield was used instead of tracking the decay of iodine into metastable xenon.

TABLE VI

Spatially averaged equilibrium concentrations in the MSRE determined after three days with constant power for the time-dependent simulations via code-to-code benchmarking with steady-state model [6].

Method	Nuclide	N_i [<i>atoms</i> · <i>cm</i> ⁻³]	d_i [%]
Spatially Resolved	¹³⁵ Xe	2.69 × 10¹⁴ <u>2.61 × 10¹⁴</u>	-
Scaled Flux	¹³⁵ Xe	2.69 × 10¹⁴ <u>2.61 × 10¹⁴</u>	0.0
Two Region [6]	¹³⁵ Xe	3.16 × 10 ¹⁴	17.3 <u>21.1</u>
Nine Region [6]	¹³⁵ Xe	2.58 × 10 ¹⁴	4.1 <u>1.1</u>
Spatially Resolved	^{135m} Xe	1.81 × 10¹² <u>1.80 × 10¹²</u>	-
Scaled Flux	^{135m} Xe	1.81 × 10¹² <u>1.80 × 10¹²</u>	0.0
Two Region [6]	^{135m} Xe	1.55 × 10 ¹²	14.1 <u>13.9</u>
Nine Region [6]	^{135m} Xe	1.57 × 10 ¹²	13.4 <u>12.8</u>
Spatially Resolved	¹³⁵ I	2.41 × 10¹⁴ <u>2.40 × 10¹⁴</u>	-
Scaled Flux	¹³⁵ I	2.41 × 10¹⁴ <u>2.40 × 10¹⁴</u>	0.0
Two Region [6]	¹³⁵ I	2.65 × 10 ¹⁴	9.7 <u>10.4</u>
Nine Region [6]	¹³⁵ I	2.35 × 10 ¹⁴	2.5 <u>2.1</u>
Spatially Resolved	¹³⁵ Te	1.04 × 10¹¹ <u>1.03 × 10¹¹</u>	-
Scaled Flux	¹³⁵ Te	1.04 × 10¹¹ <u>1.03 × 10¹¹</u>	0.0
Two Region [6]	¹³⁵ Te	9.22 × 10 ¹⁰	11.0 <u>10.5</u>
Nine Region [6]	¹³⁵ Te	9.59 × 10 ¹⁰	7.5 <u>6.9</u>
Spatially Resolved	¹³⁵ Sb	3.97 × 10⁸ <u>3.95 × 10⁸</u>	-
Scaled Flux	¹³⁵ Sb	3.97 × 10⁸ <u>3.95 × 10⁸</u>	0.0
Two Region [6]	¹³⁵ Sb	3.69 × 10 ⁸	7.1 <u>6.6</u>
Nine Region [6]	¹³⁵ Sb	3.83 × 10 ⁸	3.7 <u>3.0</u>

Table VI shows that there is good agreement between the spatially resolved model and scaled flux model for all nuclides tracked, with the largest difference being approximately 10⁻³% for ¹³⁵Xe. When comparing with the two and nine region models, in every case the nine region model has a

smaller difference. This is ~~likely~~ because the level of spatial refinement is finer in the nine region case, which brings it closer to the more spatially refined model presented in this work. ~~In the nine region model, ^{135}Sb is 3.7% smaller, ^{135}Te is 7.5% smaller, ^{135}I is 2.5% smaller, ^{135m}Xe is 13.4% smaller, and ^{135}Xe is 4.1% smaller. In the two region model, ^{135}Sb is 7.1% smaller, ^{135}Te is 11% smaller, ^{135}I is 9.7% larger, ^{135m}Xe is 14.1% smaller, and ^{135}Xe is 17.3% larger, which uses 500 spatial nodes.~~ Overall, there is reasonable agreement between the steady-state models and the current time-dependent models. The nine region model, aside from ^{135m}Xe , has differences less than 7% from the spatially resolved and scaled flux models. The larger differences in ^{135m}Xe are expected ~~, and the other nine region results are within 10%~~due to the differing treatment of that specific nuclide.

III.E. Difference Sensitivities

Based on Equations (1) and (5), the difference metric described in Section II.C can be manipulated by altering the source term, S_i ; loss term, μ_i ; and flow rate, ν . If the manipulations designed to lead to large differences result in a negligible difference, then that would prove that those scenarios, or any more conservative scenarios, do not require spatial resolution. In this work, artificial manipulation of nuclear data, such as fission yields or branching ratios, will not be considered. Instead, manipulations will be performed on properties which can be more readily controlled: the neutron flux, the chemical removal rate, and the fuel salt flow rate.

III.E.1. Flux Sensitivity

To create a gradient in the concentration through flux manipulation, the most direct way is through power oscillations in time. Specifically, a square wave can be used such that a certain portion of the fuel salt is irradiated at a high flux while another portion of fuel salt receives no irradiation. This represents the most extreme case of a spatial gradient which can be formed using flux variations. Since a realistic model would have some amount of mixing and a lesser amount of power oscillation, this provides a limiting case. If the difference metric is close to zero, then that would indicate that for power oscillation scenarios, spatial resolution is generally not necessary. A maximum difference less than 1% over tens of circulations around the primary loop would indicate that, with mixing, any realistic molten salt reactor could have power oscillations modeled during

a depletion simulation without spatial resolution.

To simplify the problem, the data is adjusted to a flow rate of 10 *cm/s* over 100 *cm*, with 50% of the salt in-core and 50% ex-core. Figure 4 shows how the concentration evolves over time. This figure normalizes the concentration at each time step by the peak concentration at that time step. This leads to the diagonal striping effect, as the peak concentration advects throughout the primary loop.

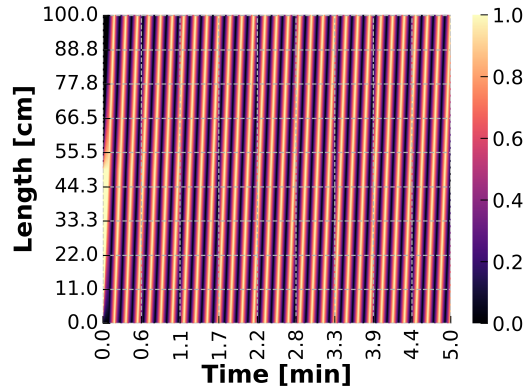


Fig. 4. Relative maximum concentration at each time step of ^{135}Xe in a toy model with 5 second in-core and ex-core residence times and a square wave power oscillation.

Figure 5 shows how the difference metric evolves over both tens and thousands of recirculations for ^{135}Xe . The five minute long simulation is sufficiently short such that the step-wise shape from the power oscillation is visible in the difference evolution via a shape with distinct steps down after each recirculation. As the fuel circulates over the first five minutes, the staircase shape trends linearly downwards, though the flat section of each portion begins to become an upwards trend. The reason for this is because of the diffusion-inducing behavior of the precursors to ^{135}Xe . Because these other nuclides decay into ^{135}Xe at different locations, the large gradient caused by the power oscillations becomes lessened. When modeled with only a single nuclide, the general trend is the same but without positive slopes.

These results show that it takes ~~approximately~~ slightly less than 5 hours, or approximately 15,000 circulations around the primary loop, before the difference metric reaches 1%. This means that, provided there is some small amount of mixing via turbulence or diffusion, the scaled flux method is sufficiently accurate to model the ^{135}Xe in the MSRE. In the case that there is no mixing, or the level of mixing is negligible, then a spatially resolved approach may be necessary

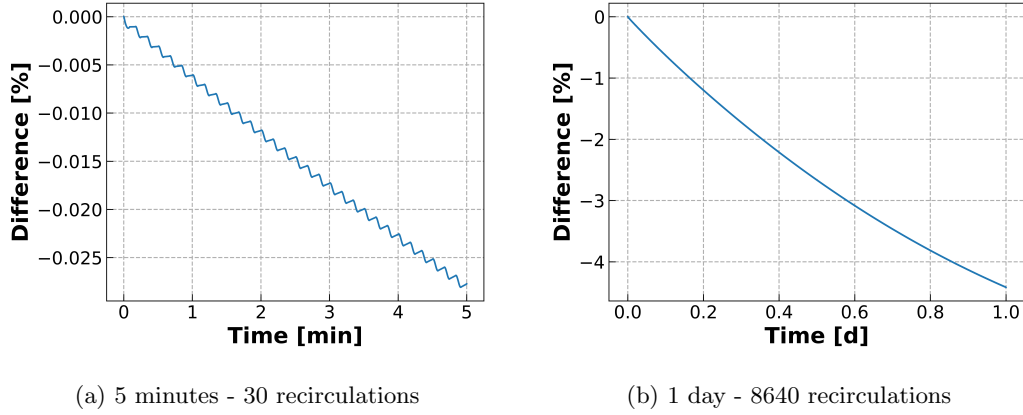


Fig. 5. ^{135}Xe difference metrics in a toy model with 5 second in-core and ex-core residence times over different periods of time.

if there are power oscillations which occur at such a frequency that leads to heterogeneity in the fuel salt.

III.E.2. Chemical Removal Sensitivity

In this work, the chemical removal is applied on each spatial node that exists in the ex-core region. The xenon chemical removal rate from the pump bowl via the off-gas system in the MSRE has been estimated in studies to be $4.067 \times 10^{-5} \text{ s}^{-1}$ [19]. ~~Similarly for the Molten Salt Breeder Reactor (MSBR), the chemical removal rate of xenon has been estimated to be 0.05 s^{-1} [20].~~ The scenario in which the largest spatial gradient would exist is one with a very large chemical removal rate in the ex-core region. A large chemical removal rate leads to the largest spatial gradient because as the chemical removal rate goes to infinity, the concentration would rapidly approach 0 in that region.

Figure 6(a) shows how the difference metric of ^{135}Xe evolves over time ~~and with varying removal rate~~. The difference ~~rate-metric~~ has a sinusoidal shape that exponentially decays, with a peak difference early on and ~~than then~~ reaching an equilibrium difference. ~~The Although Figure 6(a) shows the ^{135}Xe concentration without chemical removal, adding chemical removal does not change this shape of the difference metric evolution in time. Thus, the peak and equilibrium differences are can be~~ plotted for varying removal rate values of xenon, ~~as shown in Figure 6(b).~~

Based on the xenon removal rate in the MSRE of $4.067 \times 10^{-5} \text{ s}^{-1}$, the difference metric is negligible, meaning the scaled flux model can be used with high accuracy. ~~For the xenon removal~~

rate in the MSBR. If the MSRE had a large xenon chemical removal rate, comparable to the 0.05 s^{-1} rate in the Molten Salt Breeder Reactor (MSBR), then the difference metric would be significant [20]. For the xenon removal rate of 0.05 s^{-1} , the difference peaks at 7.5% and is at equilibrium at 0.9%. This means that the MSBR, or reactors with similar large removal rates, may need spatial resolution during depletion simulations to accurately model nuclide concentrations.

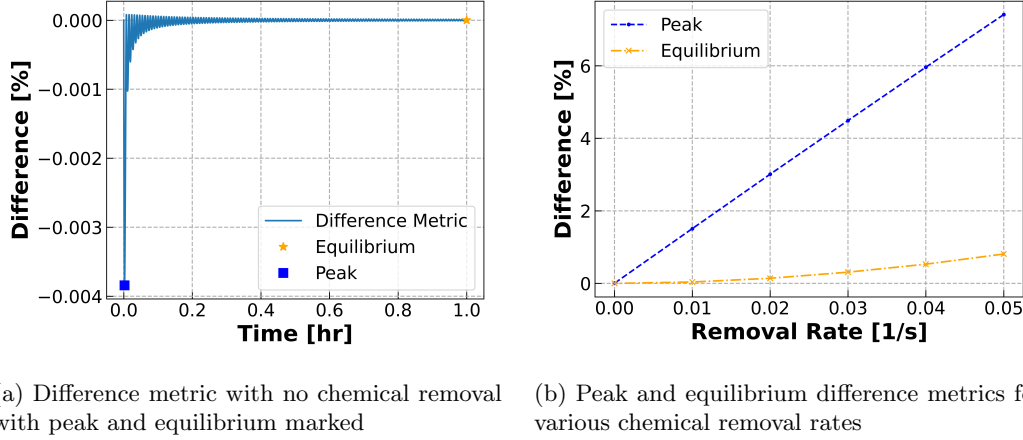


Fig. 6. ^{135}Xe chemical removal rate difference metrics in the MSRE.

III.E.3. Flow Rate Sensitivity

The flow rate is inversely proportional to the residence time of the salt in a given region. A faster flow rate, or shorter residence time in each region, is expected to lead to a smaller difference. This is because the salt would be more evenly irradiated, leading to a more homogeneous salt. As the residence time increases, the spatial gradient is expected to increase.

However, as can be seen in Equation (5), the gradient is multiplied by the flow rate. This means that there can be expected to be competing behavior from these terms. For a stationary fuel salt, the scaled flux model would no longer be correctly handling the physics, as the system is stationary and would be better modeled by the unscaled flux method. This means that, even with competing behavior, the trend should be towards an increase in the difference metric with decreasing flow rate.

Figure 7 shows how the difference metric for ^{135}Xe changes for various residence times, calculated by dividing the length of the channel by the flow rate. The peak and equilibrium difference metrics both trend upwards as the residence time increases, which aligns with expectations. This

means that for typical MSRE depletion simulations with a residence time of approximately 30 seconds, the scaled flux method has a difference metric less than 0.01% due to fuel salt advection. However, if a pump coast-down or some other loss of flow condition is part of the simulation, then the simulation will require spatial resolution to properly model the ^{135}Xe concentration.

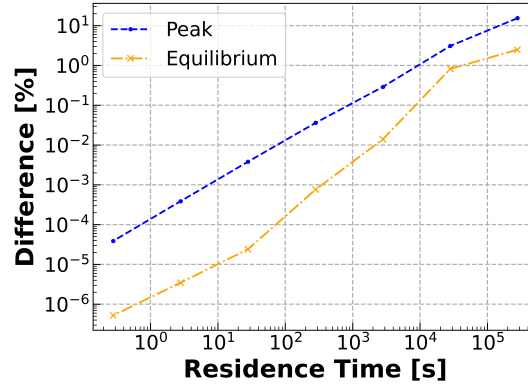


Fig. 7. Peak and equilibrium difference metrics for various primary loop residence times in the MSRE for ^{135}Xe .

IV. CONCLUSIONS

This work compares a scaled flux model, which assumes perfect mixing, and a spatially resolved model with 1D advection, which assumes no mixing, to determine the maximum error that could be expected from using the scaled flux method. ^{135}Xe was selected to be the main target of investigation as it has a large absorption cross section, and thus would be more affected by the spatial resolution than other nuclides. ~~The spatially resolved and scaled flux models offer close agreement with MCNP/ADDER, while the data from the MSRE for ^{95}Zr is in reasonable agreement.~~

The analyses in this work demonstrated the sensitivity of the difference between the scaled flux and spatially resolved models to flux, chemical removal rates, and flow rates. ~~For the flux, in general, minor~~ Minor variations in the flux can be accurately modeled using the scaled flux method. If the power is artificially oscillated to induce heterogeneity and the MSR has little to no mixing over an extended period of time, then the scaled flux method would no longer be sufficient. ~~For Small~~ chemical removal rates, ~~small removal rates~~ can be handled using the scaled flux method while large chemical removal rates, such as the removal rate of xenon in the MSBR, may require

spatial resolution. ~~For flow rates, the~~ The scaled flux method becomes increasingly inaccurate as the flow rate decreases. These results are important because they demonstrate that, for many cases, the faster and simpler scaled flux method can be used without significant loss of accuracy for modeling the concentration of nuclides. However, it is equally critical that the limitations of the method are understood and accounted for, otherwise predicted fuel compositions will have non-negligible inaccuracies.

The results in this work demonstrate the importance of both the simpler, approximate approaches and the more sophisticated, high-fidelity approaches to spatial resolution of nuclides in MSRs. The scaled flux model generally captures the physics of MSR nuclide transmutation, and for most cases offers a fast and simple approach that has negligible errors [5]. For example, the MSRE operation from January 24th, 1966 to November 7th, 1967 shows that the scaled flux model has good agreement with MCNP/ADDER and the spatially resolved model [7]. Additionally, for a steady-state solve using constant power and flow rate in the MSRE, the scaled flux method does not introduce significant error [6]. However, for some scenarios previously described, the scaled flux model becomes inaccurate, and the higher-fidelity models are necessary to accurately track nuclide concentrations [9].

Future work could involve continued verification and validation efforts for ~~isotopes~~ nuclides that exhibit in-core and ex-core loss terms due to neutron capture and chemical removal, respectively. Future work could also compare these methods to a modified-point-kinetics-like approach that treats the ex-core region with an exponential decay, offering an approach between the 0-D and 1-D approaches as shown in MacPhee [21]. Finally, future work of interest could include investigation of ~~depletion simulations of various MSRs~~ various MSR simulations in the literature and analysis on nuclide concentration difference metrics in those simulations. These nuclide concentrations affect many aspects of MSR operation, including refuel rates [22, 23], reactivity response [24], radio-toxicity of the fuel inventory [25], and more [26].

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